



CYBER SECURITY ANALYST INTERNSHIP ENGAGEMENT REPORT

Week 05: Computer Networking Fundamentals & Network Configuration

ORGANIZATION	IT SIMPLERA
REPORT TITLE	Week 05: Computer Networking Fundamentals & Network Configuration
STUDENT NAME	Dua Najam
INSTRUCTOR	Sir Talal Rehan
SUBMISSION DATE	30 July 2026

CONFIDENTIAL

FOR INTERNAL TRAINING USE ONLY

Table of Contents

1. Executive Summary.....	03
2. Learning Objectives.....	04
3. Task Documentation.....	05-17
3.1 Task 01 — Network Enumeration	
3.2 Task 02 — Service Enumeration	
3.3 Task 03 — Vulnerability Research	
3.4 Task 04 — Scan Comparison	
3.5 Task 05 — Security Analysis & Remediation Recommendations	
3.6 Task 06— Vulnerability Assessment Report	
4. Challenges Faced.....	18
5. Conclusion.....	18
6. References.....	18

1. Executive Summary

This report presents the network enumeration and vulnerability assessment performed on the assigned **Metasploitable 2** target using Nmap. Different Nmap scan types were used to identify open ports, running services, service versions, and the operating system. The collected information was analyzed to understand the target's attack surface, identify potential vulnerabilities, and recommend security improvements. All activities were conducted in an authorized lab environment for educational purposes.

Task Completion Status

Task	Description	Status
Task 01	Network Enumeration	Completed
Task 02	Service Enumeration	Completed
Task 03	Vulnerability Research	Completed
Task 04	Scan Comparison	Completed
Task 05	Security Analysis & Remediation Recommendations	Completed
Task 06	Vulnerability Assessment Report	Completed

2. Learning Objectives

- Understand the purpose of network enumeration.
- Learn how to use Nmap for network scanning.
- Identify open ports and running services.
- Detect service versions and the target operating system.
- Compare different Nmap scan types.
- Analyze the target's attack surface.
- Map discovered services to known CVEs.
- Assess security risks based on scan results.
- Recommend security hardening and remediation.
- Improve technical reporting and documentation skills.

3. Task Documentation

3.1 Task 01 — Network Enumeration

Target: Metasploitable 2

IP: 192.168.30.130

Step 01: Basic Scan

Finds open ports and basic services.

```
duaa@ubuntu-24:~$ nmap 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:04 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 1.61 seconds
```

Step 02: Version Detection

Detects service names and software versions.

```
duaa@ubuntu-24:~$ nmap -sV 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:05 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0054s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Step 03: OS Detection

Detects the Operating System.

```
duaa@ubuntu-24:~$ sudo nmap -O 192.168.30.130
[sudo] password for duaa:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:06 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:B6:2B:23 (VMware)
Device type: general purpose
```

Step 04: Aggressive Scan

Performs:

- OS Detection
- Version Detection
- Default NSE Scripts
- Traceroute

IT Simplera Red Teaming | Week 05: Computer Networking Fundamentals & Network Configuration

```
duaa@ubuntu-24:~$ sudo nmap -A 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:07 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.30.129
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
```

```
|     SSL2_RC2_128_CBC_WITH_MD5
|_   SSL2_RC4_128_WITH_MD5
|_ssl-date: 2026-07-30T15:07:53+00:00; 0s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100003   2,3,4        2049/tcp    nfs
|_  100003   2,3,4        2049/udp    nfs
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 11
|   Capabilities flags: 43564
|   Some Capabilities: ConnectWithDatabase, SupportsCompression, SupportsTransactions, Support41Auth, SwitchToSSLAfterHa
```

Step 05: Full Port Scan

Scans all 65,535 TCP ports.

```
duaa@ubuntu-24:~$ nmap -p- 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:10 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0038s latency).
Not shown: 65505 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
```

Step 06: Comparison Table

Scan Command	Purpose	Information Obtained
nmap 192.168.30.130	Basic Scan	Open ports and basic services
nmap -sV 192.168.30.130	Version Detection	Service names and software versions
sudo nmap -O 192.168.30.130	OS Detection	Operating system details
sudo nmap -A 192.168.30.130	Aggressive Scan	OS, versions, NSE scripts, traceroute
nmap -p- 192.168.30.130	Full Port Scan	All TCP ports (1–65535)

Step 07: Open Ports Table

Port	State	Service	Version
21	Open	FTP	vsftpd 2.3.4
22	Open	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23	Open	Telnet	Linux telnetd
25	Open	SMTP	Postfix smtpd
53	Open	DNS	ISC BIND 9.4.2
80	Open	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2
111	Open	RPCBind	RPC #100000
139	Open	NetBIOS-SSN	Samba smbd 3.X - 4.X
445	Open	Microsoft-DS	Samba smbd 3.0.20-Debian
512	Open	Exec	netkit-rsh rexecd
513	Open	Login	OpenBSD/Solaris rlogind
514	Open	TCP Wrapped	tcpwrapped
1099	Open	Java RMI	GNU Classpath grmiregistry
1524	Open	Bind Shell	Metasploitable root shell
2049	Open	NFS	NFS v2-4 (RPC #100003)
2121	Open	FTP	ProFTPD 1.3.1
3306	Open	MySQL	MySQL 5.0.51a-3ubuntu5

3.2 Task 02 — Service Enumeration

Port	Protocol	Service Name	Service Version	Status	Purpose
21	TCP	FTP	vsftpd 2.3.4	Open	Used to transfer files.
22	TCP	SSH	OpenSSH 4.7p1	Open	Used for secure remote login.
23	TCP	Telnet	Linux telnetd	Open	Used for remote login.
25	TCP	SMTP	Postfix smtpd	Open	Used to send emails.
53	TCP	DNS	ISC BIND 9.4.2	Open	Converts domain names to IP addresses.
80	TCP	HTTP	Apache 2.2.8	Open	Hosts websites.
111	TCP	RPCBind	RPC #100000	Open	Connects RPC services.
139	TCP	NetBIOS	Samba	Open	Shares files and printers.
445	TCP	SMB	Samba 3.0.20	Open	Shares files on the network.
512	TCP	Exec	netkit-rsh	Open	Runs remote commands.
513	TCP	Login	rlogind	Open	Allows remote login.
514	TCP	TCP Wrapped	tcpwrapped	Open	Controls service access.
1099	TCP	Java RMI	GNU Classpath	Open	Used for Java remote communication.
1524	TCP	Bind Shell	Root Shell	Open	Provides remote shell access.
2049	TCP	NFS	NFS v2–4	Open	Shares files over the network.
2121	TCP	FTP	ProFTPD 1.3.1	Open	Used to transfer files.
3306	TCP	MySQL	MySQL 5.0.51a	Open	Stores and manages databases.

3.3 Task 03 — Vulnerability Research

Service	CVE ID	Vulnerability	CVSS	Severity	Description	Fix
FTP (vsftpd 2.3.4)	CVE-2011-2523	Backdoor	10.0	Critical	This FTP version is not safe. An attacker can get access.	Update the FTP software.
SSH (OpenSSH 4.7p1)	CVE-2008-5161	Weak Security	5.8	Medium	This SSH version is old and may not be secure.	Update SSH to the latest version.
MySQL (5.0.51a)	CVE-2012-2122	Login Bypass	7.5	High	An attacker may log in without the correct password.	Update MySQL and use a strong password.

3.4 Task 04 — Scan Comparison

1. Which scan was the fastest?

Basic Nmap scan (nmap <target>) was the fastest because it only checks the default ports.

2. Which scan discovered the most information?

Aggressive scan (nmap -A <target>) discovered the most information because it found open ports, services, versions, operating system, and other details.

3. Which scan would you use in a real-world assessment and why?

I would use nmap -A because it gives more information about the target and helps identify security risks.

4. Advantages and Disadvantages

Scan	Advantages	Disadvantages
nmap	Fast and easy to use.	Gives less information.
nmap -sV	Shows service versions.	Takes more time than a basic scan.
nmap -O	Detects the operating system.	OS detection is not always correct.
nmap -A	Gives detailed information.	Slow and easy to detect by security systems.
nmap -p-	Scans all ports.	Takes the longest time.

3.5 Task 05 — Security Analysis & Remediation Recommendations

Open Port / Service	Security Risk	Attack Scenario	Risk Level	Remediation Method	Verification	Reference
21 / FTP	Old FTP version is not safe.	An attacker may get access to files.	Critical	1. Update FTP. 2. Turn off FTP if not needed. 3. Allow only trusted users.	Run nmap -sV again and check the FTP version.	NIST
22 / SSH	Old SSH version is not safe.	An attacker may try to log in.	Medium	1. Update SSH. 2. Use a strong password. 3. Disable root login.	Run nmap -sV again and check the SSH version.	CIS Benchmarks
3306 / MySQL	Old MySQL version is not safe.	An attacker may access the database.	High	1. Update MySQL. 2. Use a strong password. 3. Allow only trusted users.	Run nmap -sV again and check the MySQL version.	OWASP

3.6 Task 06 — Vulnerability Assessment Report

1. Introduction

This report presents the vulnerability assessment of the **Metasploitable 2** machine. The assessment was performed using **Nmap** to find open ports, running services, and possible security weaknesses. The purpose of this assessment was to identify security risks and suggest ways to improve system security.

2. Scope

- **Target:** Metasploitable 2
- **IP Address:** 192.168.30.130
- **Assessment Type:** Network Enumeration and Vulnerability Assessment
- **Environment:** Virtual Lab

3. Tools Used

Tool	Purpose
Nmap	Scan open ports and services
Nmap -sV	Find service versions
Nmap -O	Detect operating system
Nmap -A	Perform detailed scan
Nmap -p-	Scan all TCP ports

4. Enumeration Results

The Nmap scans found multiple open ports and services on the target system. Important services included **FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, NFS, and MySQL**. The scan also identified the operating system and software versions running on these services.

Port	State	Service	Version
21	Open	FTP	vsftpd 2.3.4
22	Open	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23	Open	Telnet	Linux telnetd
25	Open	SMTP	Postfix smtpd
53	Open	DNS	ISC BIND 9.4.2
80	Open	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2
111	Open	RPCBind	RPC #100000
139	Open	NetBIOS-SSN	Samba smbd 3.X - 4.X
445	Open	Microsoft-DS	Samba smbd 3.0.20-Debian
512	Open	Exec	netkit-rsh rexecd
513	Open	Login	OpenBSD/Solaris rlogind

Port	State	Service	Version
514	Open	TCP Wrapped	tcpwrapped
1099	Open	Java RMI	GNU Classpath grmiregistry
1524	Open	Bind Shell	Metasploitable root shell
2049	Open	NFS	NFS v2-4 (RPC #100003)
2121	Open	FTP	ProFTPD 1.3.1
3306	Open	MySQL	MySQL 5.0.51a-3ubuntu5

5. Vulnerability Findings

The assessment found several old services that may contain known security vulnerabilities.

Service	Risk	Severity
FTP	Old version	Critical
SSH	Old version	Medium
MySQL	Old version	High

Service	CVE ID	Vulnerability	CVSS	Severity	Description	Fix
FTP (vsftpd 2.3.4)	CVE-2011-2523	Backdoor	10.0	Critical	This FTP version is not safe. An attacker can get access.	Update the FTP software.
SSH (OpenSSH 4.7p1)	CVE-2008-5161	Weak Security	5.8	Medium	This SSH version is old and may not be secure.	Update SSH to the latest version.
MySQL (5.0.51a)	CVE-2012-2122	Login Bypass	7.5	High	An attacker may log in without the correct password.	Update MySQL and use a strong password.

6. Screenshots

Add the following screenshots:

- Screenshot 1 – nmap

```
duaa@ubuntu-24:~$ nmap 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:04 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 1.61 seconds
```

- Screenshot 2 – nmap -sV

```
duaa@ubuntu-24:~$ nmap -sV 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:05 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0054s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2.4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

- Screenshot 3 – nmap -O

```
duaa@ubuntu-24:~$ sudo nmap -O 192.168.30.130
[sudo] password for duaa:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:06 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:B6:2B:23 (VMware)
Device type: general purpose
```

- Screenshot 4 – nmap -A

```
duaa@ubuntu-24:~$ sudo nmap -A 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:07 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ ftp-syst:
|_   STAT:
|_   FTP server status:
|_     Connected to 192.168.30.129
|_     Logged in as ftp
|_     TYPE: ASCII
|_     No session bandwidth limit
|_     Session timeout in seconds is 300
|_     Control connection is plain text
|_     Data connections will be plain text
|_     vsFTPD 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ sslv2:
|_   SSLv2 supported
|_   ciphers:
|_     SSL2_RC4_128_EXPORT40_WITH_MD5
|_     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
```

IT Simplera Red Teaming | Week 05: Computer Networking Fundamentals & Network Configuration

```
|_ SSL2_RC2_128_CBC_WITH_MD5
|_ SSL2_RC4_128_WITH_MD5
|_ssl-date: 2026-07-30T15:07:53+00:00; 0s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp open  domain      ISC BIND 9.4.2
|_ dns-nsid:
|_ bind.version: 9.4.2
80/tcp open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp open  rpcbind     2 (RPC #100000)
|_ rpcinfo:
|_  program version    port/proto  service
|_  100003  2,3,4        2049/tcp    nfs
|_  100003  2,3,4        2049/udp    nfs
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open  exec        netkit-rsh rshcd
513/tcp open  login       OpenBSD or Solaris rlogind
514/tcp open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
|_ mysql-info:
|_  Protocol: 10
|_  Version: 5.0.51a-3ubuntu5
|_  Thread ID: 11
|_  Capabilities flags: 43564
|_  Some Capabilities: ConnectWithDatabase, SupportsCompression, SupportsTransactions, Support41Auth, SwitchToSSLAfterHa
```

- Screenshot 5 – nmap -p-

```
duaa@ubuntu-24:~$ nmap -p- 192.168.30.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-30 20:10 PKT
Nmap scan report for 192.168.30.130
Host is up (0.0038s latency).
Not shown: 65505 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
```

7. Recommendations

- Update old software to the latest version.
- Disable services that are not needed.
- Use strong passwords.
- Allow access only to trusted users.
- Use a firewall to block unnecessary ports.
- Scan the system regularly to find new vulnerabilities.

8. Conclusion

This assessment helped identify open ports, running services, and possible security risks on the Metasploitable 2 machine. The results showed that some services were outdated and could be vulnerable to attacks. By applying the recommended security measures, the system can become more secure and better protected.

4. Challenges Faced

- Understanding different Nmap scan types and their purposes.
- Some scans took longer to complete, especially full port scans.
- Interpreting service versions and scan results required careful analysis.
- Identifying relevant CVEs for discovered services required additional research.
- Organizing scan results into a professional report.

5. Conclusion

This assignment enhanced my practical knowledge of network enumeration and vulnerability assessment. By using different Nmap scan techniques, I identified open ports, running services, software versions, and the target operating system. The collected information helped me understand the attack surface and analyze potential security risks. This activity also improved my technical reporting and analytical skills required for cybersecurity roles.

6. References

- Nmap Project. (n.d.). *Nmap Reference Guide*. <https://nmap.org/docs.html>
- Nmap Project. (n.d.). *Nmap: Network Mapper*. <https://nmap.org/>
- MITRE. (n.d.). *Common Vulnerabilities and Exposures (CVE)*. <https://cve.mitre.org/>
- NIST. (n.d.). *National Vulnerability Database (NVD)*. <https://nvd.nist.gov/>